PROBLEM SET 4 DUE 4.12

1. Let $F : \mathscr{C} \to \mathscr{D}$ be a functor where $\mathscr{C}, \mathscr{D}$ are categories. It is easy to show that F preserves isomorphisms. That is, if $f : X \to Y$ is an isomorphism in $\mathscr{C}$, then $F(f) : F(X) \to F(Y)$ is an isomorphism in $\mathscr{D}$. In particular, F preserves isomorphism relations of objects. That is, if $X \cong Y$ in $\mathscr{C}$, then $F(X) \cong F(Y)$ in $\mathscr{D}$.

We say that F **reflects isomorphisms** if the following holds for any morphism $f : X \to Y$ in $\mathscr{C}$:

$$F(f) \text{ is an isomorphism in } \mathscr{D} \quad \Longrightarrow \quad f \text{ is an isomorphism in } \mathscr{C}.$$

For example, the forgetful functors from **Grp/Ring/**A**-Mod** to **Set** reflect isomorphisms, whereas the forgetful functor from **Top** to **Set** does not. Also, for any ring map $A \to B$, the restriction-of-scalar functor $|_A : B\textbf{-Mod} \to A\textbf{-Mod}$ reflects isomorphisms.

We say that F **reflects isomorphisms relations** if, for any objects X, Y of $\mathscr{C}$, $F(X) \cong F(Y)$ in $\mathscr{D}$ implies $X \cong Y$ in $\mathscr{C}$.

(1) Find an example of a functor which reflects isomorphisms but does not reflect isomorphism relations. Find an example of a functor which reflects isomorphism relations but does not reflect isomorphisms.

(2) Show that if F is fully faithful, then F reflects isomorphisms and reflects isomorphism relations.

2. Let $F, G : \mathscr{A} \to \mathscr{B}$ be functors and let $\alpha = (\alpha_X)_X : F \Rightarrow G$ be a natural transformation. Show that α is an isomorphism in the functor category $[\mathscr{A}, \mathscr{B}]$ if and only if for any $X \in \mathscr{A}$, α_X is an isomorphism in $\mathscr{B}$.

3. Let $F : \mathscr{A} \to \mathscr{B}$; $G_1, G_2 : \mathscr{B} \to \mathscr{C}$; $H : \mathscr{C} \to \mathscr{D}$ be functors where $\mathscr{A}, \mathscr{B}, \mathscr{C}, \mathscr{D}$ are categories, and let $\alpha : G_1 \Rightarrow G_2$ be a natural transformation.

(1) Show that
$$\alpha F \coloneqq \Big\{\alpha_{F(X)} : G_1(F(X)) \to G_2(F(X))\Big\}_{X \in \mathscr{A}}$$
is a natural transformation from G_1F to G_2F.

(2) Show that
$$H\alpha \coloneqq \Big\{H(\alpha_Y) : H(G_1(Y)) \to H(G_2(Y))\Big\}_{Y \in \mathscr{B}}$$
is a natural transformation from HG_1 to HG_2.

4. Let (X_1, u_1) and (X_2, u_2) be two representations of a functor $F : \mathscr{C} \to \textbf{Set}$. Let $f : X_1 \to X_2$ be the unique morphism such that $F(f)(u_1) = u_2$, and let $g : X_2 \to X_1$ be the unique morphism such that $F(g)(u_2) = u_1$. Show that $gf = \mathrm{id}_{X_1}$ and $fg = \mathrm{id}_{X_2}$.

5. Find a representation for each of the following functors:

(1) the forgetful functor from **Grp** to **Set**;
(2) the forgetful functor from **Ring** to **Set**;
(3) the identity functor from **Set** to **Set**.